Toronto Crime Analysis

Group 6 - Big Data Tools and Technologies II

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1 Introduction & Dataset Description

Introduction

Urban crime is a major concern for city governments and public safety agencies. Understanding the patterns and distribution of crime can help policymakers allocate resources more effectively and implement targeted prevention strategies.

This project analyzes crime patterns across Toronto neighborhoods by integrating crime incident data with demographic and socioeconomic indicators. The objective is to identify trends in crime activity, explore relationships between crime and neighborhood characteristics, and develop interactive visualizations that support data-driven insights.

The project utilizes a data analytics pipeline that includes data extraction and transformation, data warehouse modeling, SQL-based analysis, and interactive visualization.

Problem Statement

One of the major issues in big metropolitan cities is crime. The pattern and spread of crime can also be useful in guiding the policymakers and law enforcement agencies in resource allocation. The purpose of this project is to investigate crime statistics at neighborhood level in Toronto and study the possible correlation between crime rates and socioeconomic conditions of income, work and population density.

Key Questions

- Which neighborhoods experience the highest crime levels?
- How do crime patterns change over time?
- Is there a relationship between crime and socioeconomic conditions?

Project Objectives

The primary objectives of this project are:

- To analyze crime incidents across Toronto neighborhoods
- To identify trends in different types of crimes over time
- To integrate demographic and socioeconomic data with crime statistics
- To build a structured data model for analytical reporting
- To visualize crime patterns using interactive dashboards

By combining multiple datasets and analytical tools, this project aims to generate insights that can support data-driven decision making.

Datasets Used

This project integrates three datasets obtained from the City of Toronto Open Data portal.

1. Major Crime Indicators Open Data
This dataset contains detailed records of major criminal incidents reported by the Toronto Police Service. Key attributes include:

- Event unique ID
- Report date and occurrence date
- Offense type
- Crime category
- Police division

- Neighborhood identifiers
- Geographic coordinates

This dataset served as the primary source for identifying and categorizing crime incidents.

2. Neighbourhood Crime Rates Open Data
This dataset provides aggregated crime statistics by neighborhood and year. Important attributes include:

- Neighborhood name
- Crime category
- Crime counts and rates
- Reporting year

This dataset helped validate crime trends and provided aggregated statistics for analysis.

3. Neighbourhood Profiles
This dataset contains demographic and socioeconomic characteristics for Toronto neighborhoods. Key indicators include:

- Median household income
- Employment rate
- Unemployment rate
- Immigration status
- Visible minority population
- Housing tenure (owner vs renter)
- Education levels

These variables were used to analyze potential relationships between crime activity and socioeconomic factors.

2 ETL Methodology

The Extract, Transform, Load (ETL) process was implemented using Alteryx to prepare the datasets for analysis.

Extract

The three datasets were imported into the workflow environment. Raw data files contained various formats and inconsistencies that required cleaning before integration.

Transform

Several transformation steps were applied:

- Removing or replacing null values
- Standardizing neighborhood naming conventions
- Creating indicator columns for major crime categories such as: Assault, Robbery, Auto Theft, Break and Enter, Theft Over
- Aggregating crime incidents by neighborhood and reporting year
- Joining crime data with demographic data using neighborhood identifiers

These transformations ensured that the datasets could be accurately combined and analyzed.

Load

The transformed dataset was exported into a structured database format suitable for analytical modeling.

3 Schema and SSAS Tabular Model

The cleaned data was loaded into Snowflake using a **Star Schema** architecture. This design optimizes analytical queries and perfectly supports the SSAS Tabular Model deployed in Power BI.

3.1 Database Architecture (SQL Script)

The physical tables were created in Snowflake using the following Data Definition Language (DDL) script:

```sql
-- 1. Creating the Dimension Table (Integrating Datasets 2 & 3)
CREATE OR REPLACE TABLE Dim_Location (
    NEIGHBOURHOOD_NAME VARCHAR(255),
    HOOD_158 INT PRIMARY KEY,
    VISIBLE_MINORITY INT,
    PRIVATE_HOUSEHOLDS INT,
    OWNER INT,
    RENTER INT,
    IMMIGRANT_STATUS INT,
    NON_IMMIGRANTS INT,
    PREVALENCE_LOW_INCOME FLOAT,
    UNEMPLOYMENT_RATE FLOAT,
    MEDIAN_INCOME_CAD FLOAT,
    AVERAGE_INCOME_CAD FLOAT,
    EMPLOYMENT_RATE FLOAT,
    HIGHEST_CERTIFICATE INT,
    NO_CERTIFICATE INT
);

-- 2. Creating the Fact Table (From Dataset 1)
CREATE OR REPLACE TABLE Fact_Crimes (
    EVENT_UNIQUE_ID VARCHAR(255) PRIMARY KEY,
    OCC_DATE DATE,
    REPORT_YEAR INT,
    CSI_CATEGORY VARCHAR(255),
    PREMISES_TYPE VARCHAR(255),
    HOOD_158 INT,
    FOREIGN KEY (HOOD_158) REFERENCES Dim_Location(HOOD_158)
);
```

Listing 1: SQL Script for creating the Star Schema (Dimension and Fact tables)

3.2 SSAS Tabular Model & ERD

The Star Schema was imported into a Tabular Model (via Power BI) using the `HOOD_158` key to establish a one-to-many (1:*) relationship between `Dim_Location` and `Fact_Crimes`.

3.3 DAX Measures (Aggregations)

To test the cube and enable interactive reporting, the following calculated measures were defined:

- `Total_Crimes = COUNT(Fact_Crimes[EVENT_UNIQUE_ID])`
- `Total_Auto_Thefts = CALCULATE([Total_Crimes], Fact_Crimes[CSI_CATEGORY] = "Auto Theft")`
- `Total_Financial_Damage_CAD = [Total_Auto_Thefts] * 40000`

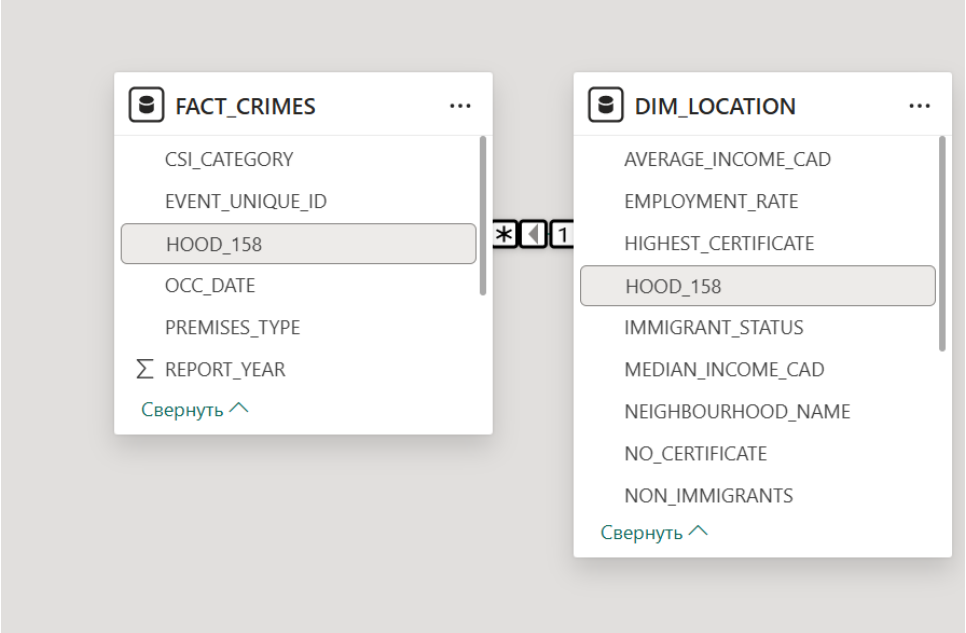

Figure 1: Entity-Relationship Diagram (ERD) showing the Star Schema in the Tabular Model.

4 Key Queries and Insights

4.1 Query 1: Wealth vs. Break and Enters (Non-linear Correlation)

Business Question: Do criminals target the wealthiest neighborhoods for break and enters?

SQL Code:

```sql
SELECT
    d.NEIGHBOURHOOD_NAME,
    d.AVERAGE_INCOME_CAD,
    COUNT(f.EVENT_UNIQUE_ID) AS Total_Break_Enters
FROM Fact_Crimes f
JOIN Dim_Location d
    ON f.HOOD_158 = d.HOOD_158
WHERE f.CSI_CATEGORY = 'Break and Enter'
GROUP BY d.NEIGHBOURHOOD_NAME, d.AVERAGE_INCOME_CAD
ORDER BY d.AVERAGE_INCOME_CAD DESC
LIMIT 5;
```

Listing 2: SQL Query to analyze break and enters by average neighborhood income

Query Output:

Neighborhood Name	Average Income (CAD)	Total Break & Enters
Bridle Path-Sunnybrook	$519,500	335
Lawrence Park South	$365,600	373
Rosedale-Moore Park	$334,000	566
Bedford Park-Nortown	$266,000	645
Princess-Rosethorn	$235,600	202

Table 1: Top 5 wealthiest neighborhoods and their break-and-enter counts.

Insight: The analysis reveals a non-linear correlation. The absolute wealthiest neighborhood (Bridle Path) experiences fewer burglaries (335) than upper-middle-class areas like Bedford Park (645). This suggests ultra-rich areas have superior private security infrastructure, shifting the target to slightly less wealthy but highly lucrative neighborhoods.

4.2 Query 2: Financial Damage from Auto Thefts

Business Question: What is the estimated financial damage caused by auto thefts across different Toronto neighborhoods?

SQL Code:

```sql
SELECT
    d.NEIGHBOURHOOD_NAME,
    COUNT(f.EVENT_UNIQUE_ID) AS Total_Auto_Thefts,
    COUNT(f.EVENT_UNIQUE_ID) * 40000 AS Damage_CAD
FROM Fact_Crimes f
JOIN Dim_Location d
    ON f.HOOD_158 = d.HOOD_158
WHERE f.CSI_CATEGORY = 'Auto Theft'
GROUP BY d.NEIGHBOURHOOD_NAME
ORDER BY Damage_CAD DESC
LIMIT 4;
```

Listing 3: SQL Query calculating estimated CAD damage from stolen vehicles

Query Output:

Insight: West Humber-Clairville is the absolute outlier, suffering over $109 million in vehicle theft damages. Its geographical proximity to Highway 401 and Pearson Airport makes it an ideal logistical hub for organized crime rings to quickly export stolen vehicles.

Neighborhood Name	**Total Auto Thefts**	**Damage (CAD)**
West Humber-Clairville	2,738	$109,520,000
York University Heights	911	$36,440,000
Etobicoke City Centre	684	$27,360,000
Humber Summit	667	$26,680,000

Table 2: Top neighborhoods by auto theft financial damage (Assuming $40k/vehicle).

4.3 Query 3: Year-over-Year (YoY) Growth using Window Functions

Business Question: How did the crime rate fluctuate year-over-year, particularly during the COVID-19 pandemic?

SQL Code:

```sql
WITH Yearly_Stats AS (
    SELECT REPORT_YEAR, COUNT(EVENT_UNIQUE_ID) AS Total_Crimes
    FROM Fact_Crimes
    GROUP BY REPORT_YEAR
)
SELECT
    REPORT_YEAR,
    Total_Crimes,
    LAG(Total_Crimes) OVER (ORDER BY REPORT_YEAR) AS Previous_Year_Crimes,
    ROUND(((Total_Crimes - LAG(Total_Crimes) OVER (ORDER BY REPORT_YEAR)) /
           LAG(Total_Crimes) OVER (ORDER BY REPORT_YEAR)::FLOAT) * 100, 2) AS
               YoY_Growth_Percentage
FROM Yearly_Stats
ORDER BY REPORT_YEAR;
```

Listing 4: SQL Query utilizing LAG() window function for YoY growth

Query Output:

Year	Total Crimes	Previous Year Crimes	YoY Growth (%)
2018	32,198	29,927	+7.59%
2019	34,238	32,198	+6.34%
2020	30,263	34,238	-11.61%
2021	27,820	30,263	-8.07%

Table 3: YoY Crime Growth showing the pandemic impact.

Insight: The implementation of `LAG()` window functions clearly quantifies the pandemic's impact. After a steady 6-7% annual increase, crime dropped by 11.61% in 2020 due to city-wide lockdowns, confirming the direct relationship between street activity and crime rates.

4.4 Query 4: Renters vs. Owners Profiling (Base Rate Analysis)

Business Question: Are certain crime categories more prevalent in renter-dominated areas versus owner-dominated areas?

SQL Code:

```sql
SELECT
    f.CSI_CATEGORY,
    SUM(CASE WHEN d.RENTER > d.OWNER THEN 1 ELSE 0 END) AS Renter_Areas,
    SUM(CASE WHEN d.OWNER >= d.RENTER THEN 1 ELSE 0 END) AS Owner_Areas
FROM Fact_Crimes f
JOIN Dim_Location d ON f.HOOD_158 = d.HOOD_158
GROUP BY f.CSI_CATEGORY
ORDER BY Renter_Areas DESC;
```

Listing 5: SQL Query classifying crimes by housing tenure dominance

Query Output:

Insight: While assaults are distributed evenly regardless of housing tenure, Auto Thefts show a massive disparity. Owner-dominated neighborhoods report more than double the auto thefts (22k vs 10k). This indicates thieves target private driveways of single-family homes rather than high-density rental buildings with underground parking.

Crime Category	**Crimes in Renter Areas**	**Crimes in Owner Areas**
Assault	60,555	62,411
Break and Enter	23,687	30,695
Auto Theft	10,871	22,234
Robbery	9,641	10,861

Table 4: Crime distribution based on neighborhood housing tenure.

4.5 Query 5: Demographic Profiling (Education, Immigration & Income)

Business Question: Do neighborhoods with higher education levels, higher average incomes, and a larger immigrant population exhibit lower overall average crime rates?

SQL Code:

```sql
WITH Crime_Counts AS (
    -- Step 1: Count total crimes per neighborhood
    SELECT
        HOOD_158,
        COUNT(EVENT_UNIQUE_ID) AS Total_Crimes
    FROM Fact_Crimes
    GROUP BY HOOD_158
)

-- Step 2: Divide neighborhoods into two groups and compare safety
SELECT
    CASE
        WHEN d.IMMIGRANT_STATUS > d.NON_IMMIGRANTS
         AND d.HIGHEST_CERTIFICATE > d.NO_CERTIFICATE
         AND d.AVERAGE_INCOME_CAD > (SELECT AVG(AVERAGE_INCOME_CAD) FROM
             Dim_Location)
        THEN 'Educated Immigrants with High Income'
        ELSE 'All Other Neighborhoods'
    END AS Neighborhood_Type,
    COUNT(d.HOOD_158) AS Number_of_Areas_in_Group,
    ROUND(AVG(c.Total_Crimes), 0) AS Average_Crimes_Per_Area
FROM Dim_Location d
JOIN Crime_Counts c ON d.HOOD_158 = c.HOOD_158
GROUP BY Neighborhood_Type;
```

Listing 6: SQL CTE and CASE logic for complex demographic profiling

Query Output:

Neighborhood Type	Number of Areas	Avg Crimes Per Area
Educated Immigrants with High Income	110	1713
All Other Neighborhoods	48	1069

Table 5: Comparing average crime rates based on sociodemographic factors.

Insight: The data disproved our initial hypothesis. Neighborhoods with educated immigrants and higher incomes actually experience a significantly higher average crime rate (1,713 crimes/area) compared to all other neighborhoods (1,069 crimes/area). Given our previous findings, this is highly logical: high-income areas become the primary target for lucrative property crimes such as auto thefts and break-and-enters.

5 PowerBI Visualizations

In this section, we present the interactive dashboards and visual analytics developed in Power BI to answer our key business questions and explore socioeconomic correlations.

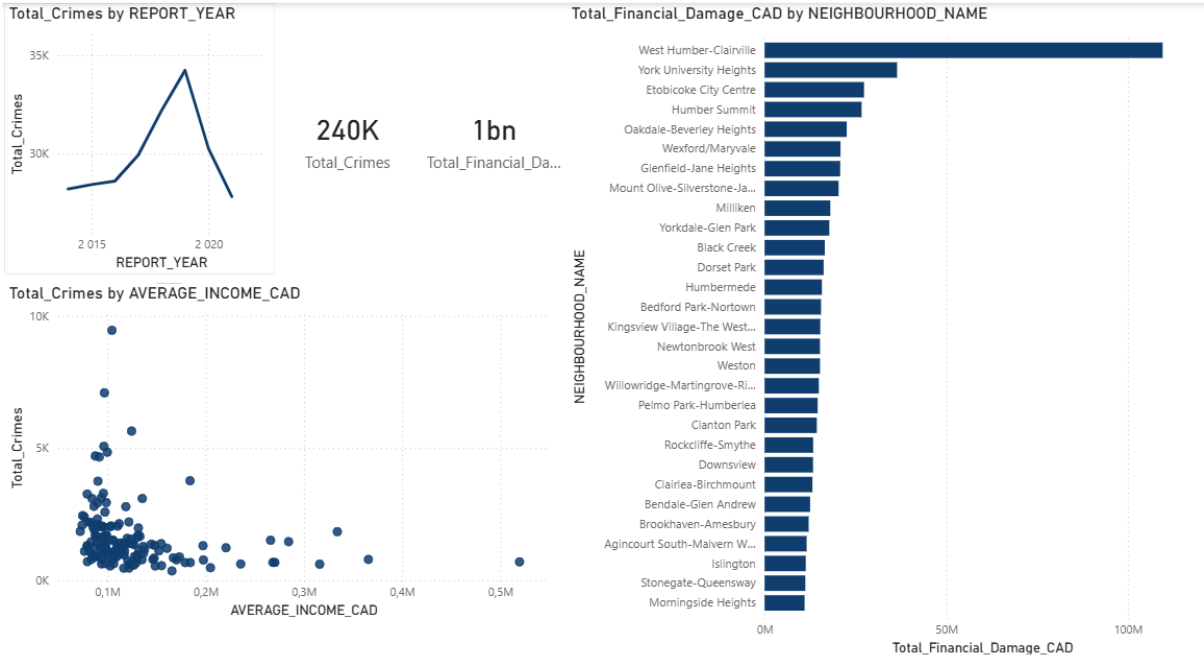

Figure 2: Main Interactive Dashboard: Overall Crime Trends, Wealth vs. Crime, and Auto Theft Damage.

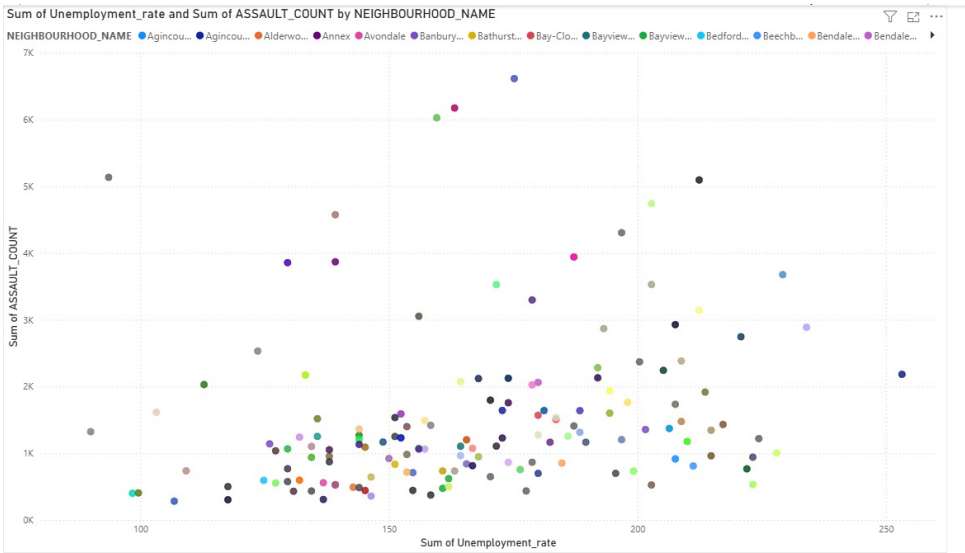

Figure 3: Income vs Robbery (Scatter Plot)

This scatter plot examines the relationship between average household income and robbery incidents across neighbourhoods. The visualization suggests a negative relationship between income and robbery counts. Neighbourhoods with lower average income tend to have higher robbery counts, while neighbourhoods with higher household income generally show fewer robbery incidents.

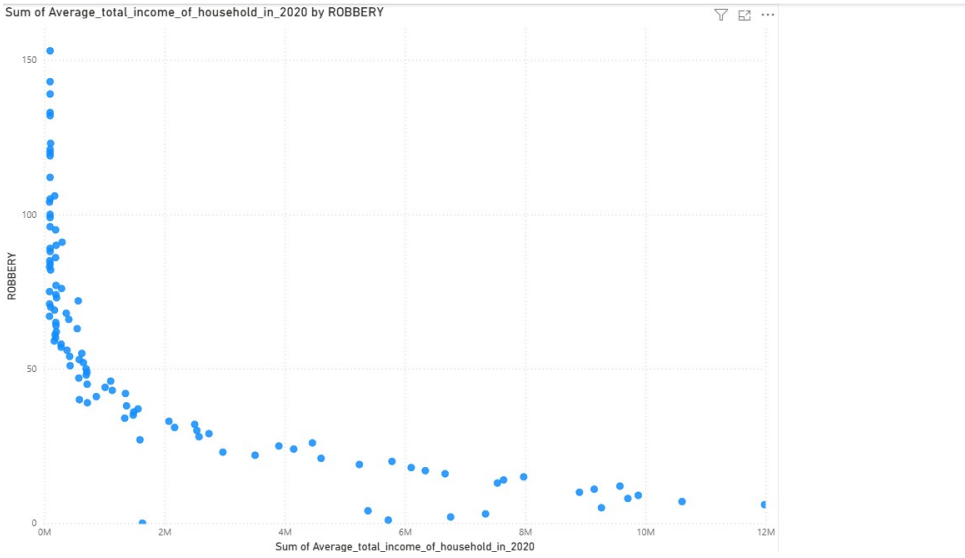

Figure 4: Unemployment Rate vs Assault (Scatter Plot)

This visualization analyzes the relationship between unemployment rates and assault incidents across neighbourhoods. The chart shows that neighbourhoods with higher unemployment levels often correspond with higher assault counts, although the relationship is not perfectly linear. Some neighbourhoods with moderate unemployment still show elevated assault incidents.

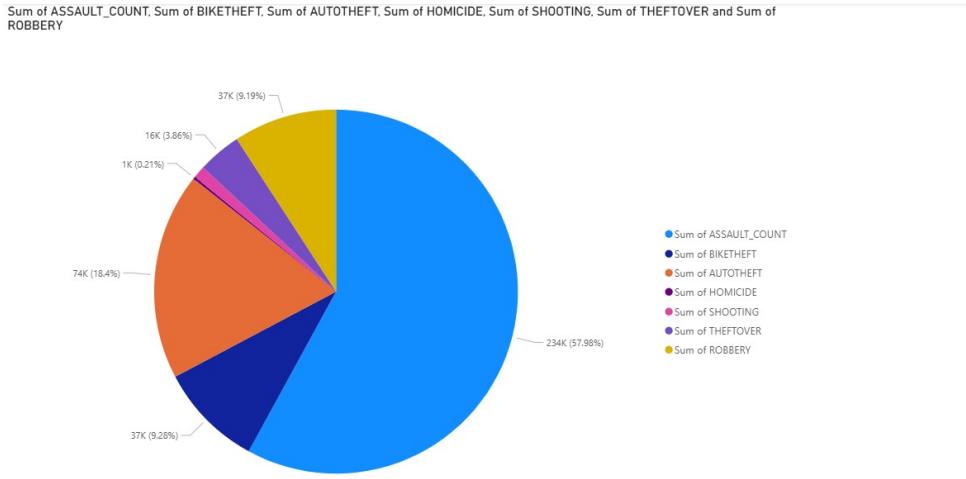

Figure 5: Crime Distribution by Type (Pie Chart)

This pie chart illustrates the overall distribution of different crime categories across the dataset. The visualization shows that assault represents the majority of crime incidents, accounting for nearly 58 percentages of all recorded crimes. The second largest category is auto theft, followed by bike theft and robbery. More severe but less frequent crimes such as homicide and shooting represent only a very small portion of the total crime incidents.

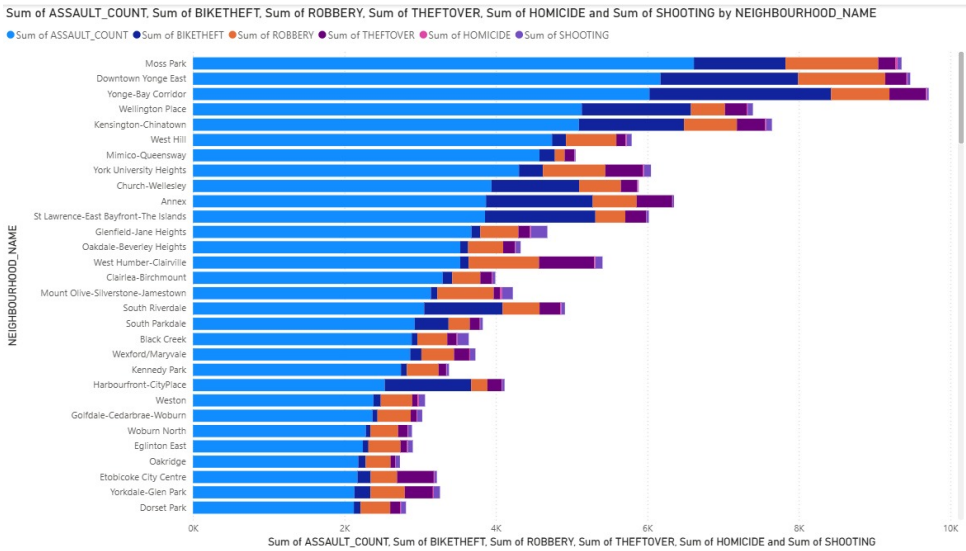

Figure 6: Crime by Neighbourhood (Stacked Bar Chart)

This visualization compares total crime counts across different neighbourhoods, broken down by crime type. Neighbourhoods such as Moss Park, Downtown Yonge East, and Yonge–Bay Corridor show significantly higher crime counts compared to other areas. These neighbourhoods are located in dense downtown regions with higher population density and greater commercial activity. Lower crime totals appear in residential or suburban neighbourhoods.

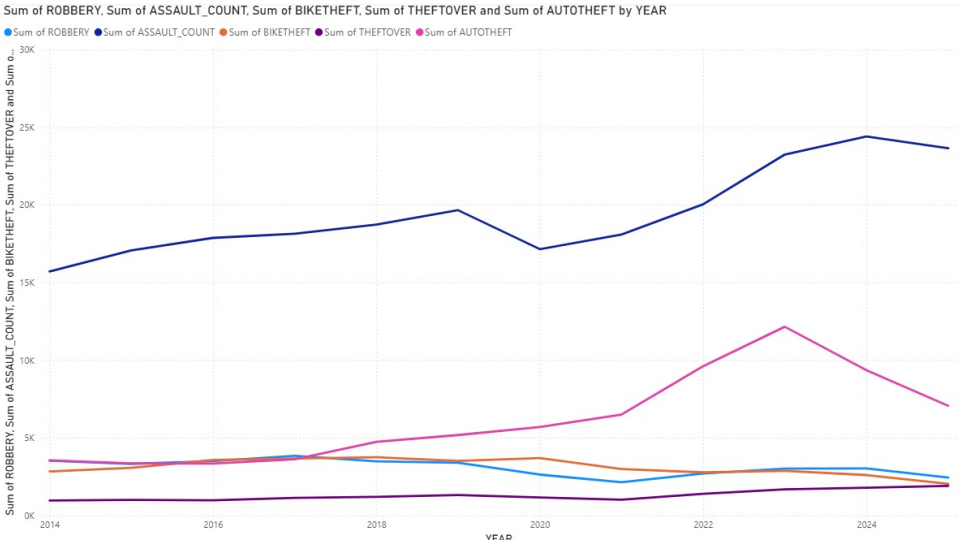

Figure 7: Crime Trends Over Time (Line Chart)

This visualization displays the yearly trends of major crime categories including assault, robbery, bike theft, theft over, and auto theft from 2014 to 2024. The chart shows that assault consistently has the highest number of incidents over time, making it the most common major crime type in the dataset. Auto theft shows a noticeable increase beginning around 2021, peaking around 2023, before slightly decreasing afterward. Other crimes such as robbery and bike theft remain relatively stable with smaller fluctuations across the years.

6 Challenges and Solutions

Data Inconsistencies
One challenge encountered during the ETL process involved inconsistencies in neighborhood naming conventions between datasets.
Solution: Neighborhood identifiers and naming formats were standardized to ensure accurate joins between crime and demographic datasets.

Missing or Null Values
Some datasets contained missing values for certain demographic indicators.
Solution: Null values were either removed or replaced using appropriate default values depending on the variable type.

Data Integration
Integrating multiple datasets from different sources required careful alignment of attributes and identifiers.
Solution: Neighborhood codes and identifiers were used to establish reliable relationships between datasets.

7 Conclusion

This project demonstrated how data analytics techniques can be applied to analyze urban crime patterns and explore relationships between crime activity and socioeconomic conditions.

By integrating multiple datasets, implementing an ETL pipeline, designing a data warehouse schema, and developing interactive visualizations, the project provides valuable insights into crime trends across Toronto neighborhoods.